Topic 2 Review Test · IEP · ANSWER KEY

Algebra I · Module 2, Topic 2: Transforming and Comparing Linear Functions · Monday, Nov 2

Directions: This review has the same kinds of questions as the Topic 2 test. Read each question carefully. Use the steps and tips. Fill in each blank. For multiple choice, choose **one** answer unless it says **Select TWO**. Finish Part 1 first; Part 2 is extra practice. One question is on each page. Take your time. You may use a calculator and the reference box. Ask your teacher if you need a break or a question read aloud.

Reference box

Slope: **$m = (y_2 - y_1) \div (x_2 - x_1)$** · Slope-intercept form: **$y = mx + b$** · Point-slope form: **$y - y_1 = m(x - x_1)$**

Parallel lines: same slope. **Perpendicular** lines: negative reciprocal slopes (2 and $-\frac{1}{2}$).

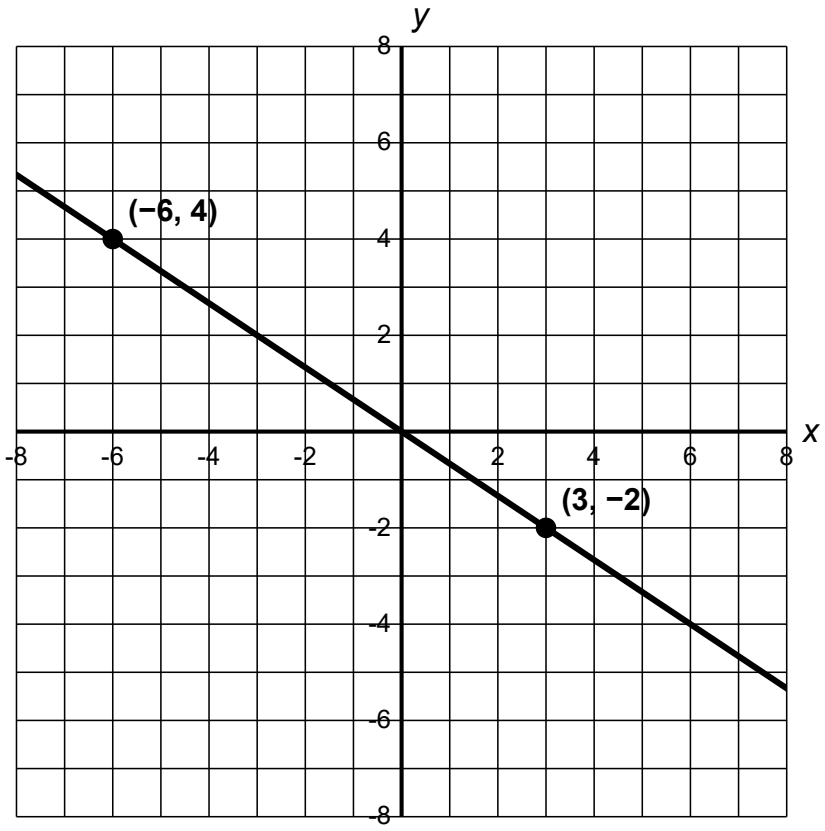
f(x) + k: up k · **f(x) − k**: down k · **f(x − h)**: right h · **f(x + h)**: left h · **a·f(x)** or **f(ax)**: steeper or flatter

Domain = inputs (x). **Range** = outputs (y). **Zero** = where the line crosses the x-axis.

Part 1 · Just like the test (Questions 1–10)

1. The graph of linear function m is shown. Given $f(x) = x$ and $m(x) = af(x)$, find a.

> **Tip:** In $m(x) = a \cdot x$, the number a is the **slope** of the line.



Step 1: slope = $\frac{-2 - 4}{3 - (-6)}$ = $\frac{-6}{9}$

Step 2: Simplify. a = **−2/3**

2. A bounce house costs $75, plus $20 per hour. It can be rented for at most 5 hours. What is the **range**?

> **Tip: Range** = the costs (the y-values).

Step 1: Cost for 0 hours: 75 + 20(0) = **75**

Step 2: Cost for 5 hours: 75 + 20(5) = **175**

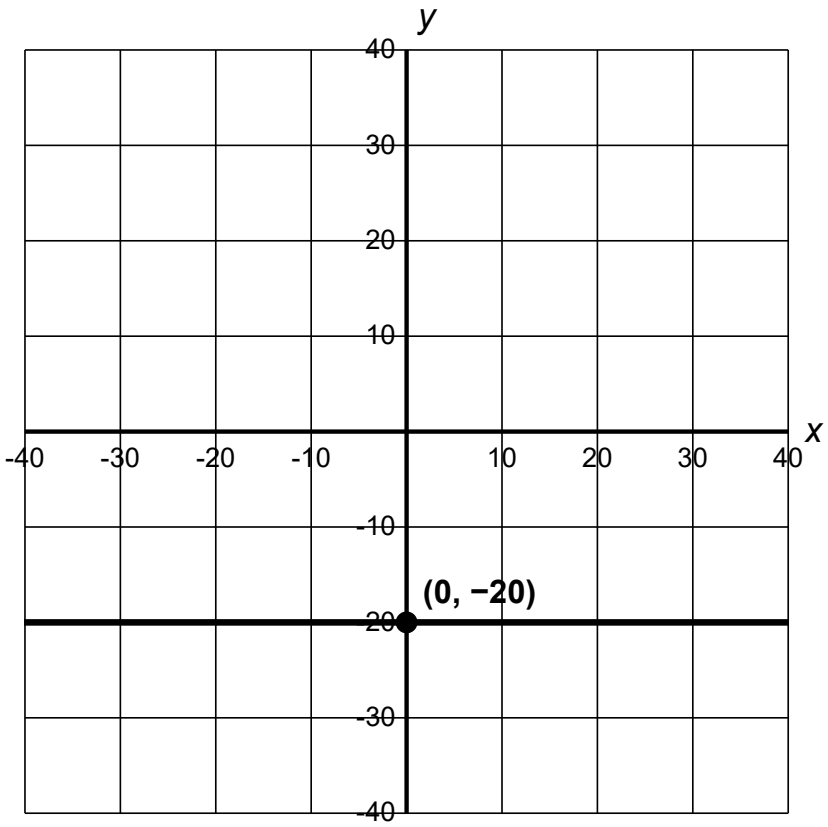
A. $0 \le x \le 5$

B. **$75 \le y \le 175$**

C. $20 \le y \le 175$

3. What are the **equation and slope** of the line shown on the grid?

> **Tip:** A flat (horizontal) line is *y* = a number. Its slope is **zero**.



A. ***y* = −20; slope is zero**

B. *x* = −20; slope is undefined

C. *y* = −20; slope is undefined

4. Marcus graphed $f(x)$ and $g(x) = f(x - 3)$. How does the graph of f move to make g?

> **Tip:** A number **inside** the parentheses moves the graph left or right, the **opposite** way of the sign: $-3 \rightarrow$ right.

A. up 3 units

B. **right 3 units**

C. left 3 units

5. Move $f(x)$ in the table **2 units right**. What is the new ***y*-intercept**?

> **Tip:** Right 2 means add 2 to each x. The y-intercept is the point where $x = 0$.

x	$f(x)$
−2	−7
−1	−4
0	−1
1	2
2	5
3	8

Step 1: The first row is the point (−2, **−7**).

Step 2: Move it right 2: (−2 + 2, **−7**) = (0, **−7**)

Step 3: New y-intercept = **−7**

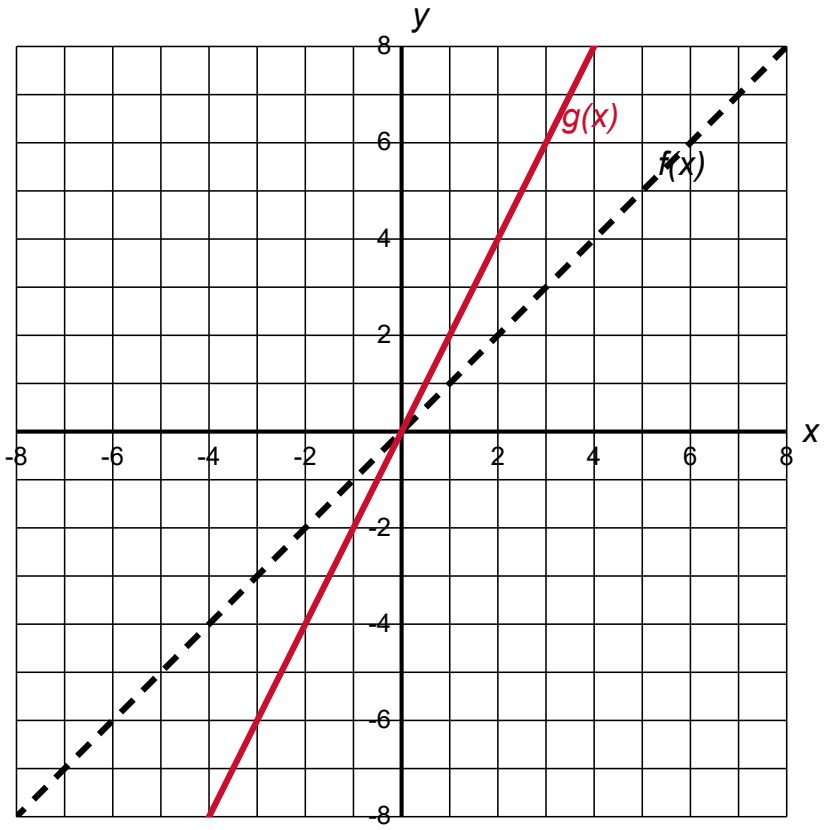
6. $f(x) = x$ (dashed) and $g(x) = f(2x)$. Graph $g(x)$.

> **Tip:** $f(2x)$ means put $2x$ in for x: $g(x) = 2x$.

Step 1: Fill in the table for $g(x) = 2x$.

x	0	1	2	3
$g(x)$	**0**	**2**	**4**	**6**

Step 2: Plot the points and connect them with a line.



7. Write the equation in **slope-intercept form** of the line that crosses the x-axis at -6 and is **perpendicular** to $y = \frac{3}{2}x - 1$.

> **Tip: Perpendicular** slope: flip the fraction and change the sign.

Step 1: The slope of $y = \frac{3}{2}x - 1$ is **3/2**.

Step 2: The perpendicular slope is **−2/3**.

Step 3: Crossing the x-axis at -6 is the point (**−6** , **0**).

Step 4: $0 = -\frac{2}{3}(-6) + b$, so $b =$ **−4**

Step 5: $y =$ **−2/3** $x +$ **(−4)**

8. Write the equation in **slope-intercept form** of the line through (5, 0) that is **parallel** to $y = -2.4x + 1.6$.

> **Tip: Parallel** lines have the **same** slope.

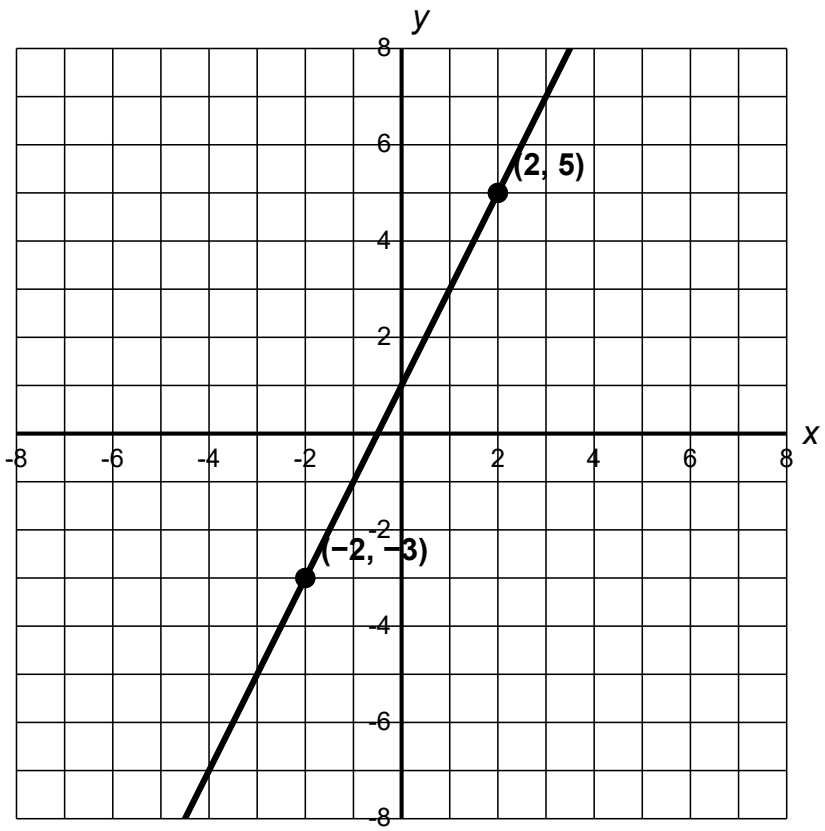
Step 1: Slope = **−2.4**

Step 2: $0 = -2.4(5) + b \rightarrow 0 =$ **−12** $+ b \rightarrow b =$ **12**

Step 3: $y =$ **−2.4** $x +$ **12**

9. Write an equation in **point-slope form** for the graph.

> **Tip:** Point-slope form: $y - y_1 = m(x - x_1)$. Use the point (2, 5).

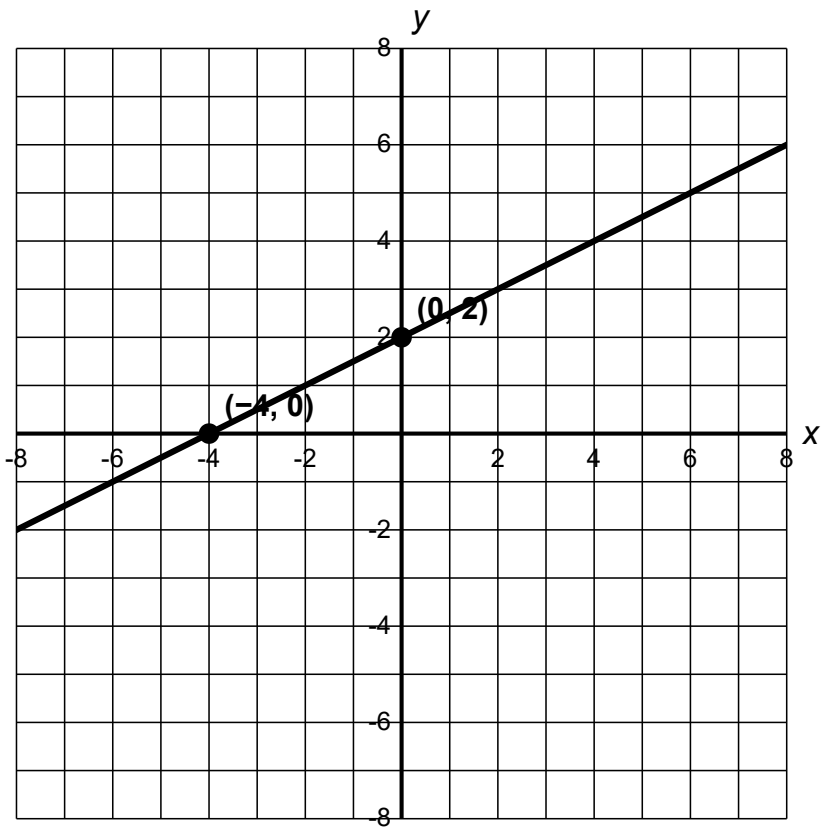


Step 1: slope = $\frac{5-(-3)}{2-(-2)} = \frac{8}{4} = 2$

Step 2: $y -$ 5 $= 2\,(x -$ 2 $)$

10. The graph of a linear function is shown. Which statements are true? **Select TWO correct answers.**

> **Tip:** The **zero** is where the line crosses the x-axis. **Increasing** means it goes up from left to right.



A. **The graph is increasing.**

B. The slope of the line is 2.

C. **The zero of the function is −4.**

D. The zero of the function is 2.

Part 2 · Extra practice (Questions 11–14)

11. $f(x) = x - 1$, $g(x) = f(2x)$, and $h(x) = 2f(x)$. Compare g and h.

> **Tip:** $f(2x)$: put $2x$ in for x. $2f(x)$: multiply the whole function by 2.

Step 1: $g(x) = 2x -$ **1**

Step 2: $h(x) = 2(x - 1) = 2x -$ **2**

Step 3: Same slope? **yes** / **no** (circle one) Same y-intercept? **yes** / **no** (circle one)

12. Which pair of equations represents **perpendicular** lines?

> **Tip:** Multiply the two slopes. If you get **−1**, the lines are perpendicular.

A. $y = 5x + 1$ and $y = -5x + 2$

B. $y = -\frac{1}{3}x + 4$ and $y = 3x - 2$

C. $y = 2x + 3$ and $y = 2x - 3$

13. $f(x) = 2x + 1$ and $g(x) = f(x) - 4$. Write $g(x)$ and describe the move.

> **Tip:** A number **outside** the function moves the graph up or down.

Step 1: $g(x) = 2x + 1 - 4 = 2x -$ **3**

Step 2: The graph moves **up** / **down** (circle one) **4** units.

14. A 300-gallon tank drains 15 gallons per minute until it is empty. What is the **domain**?

> **Tip: Domain** = the minutes (the x-values).

Step 1: Minutes to empty: 300 ÷ 15 = **20**

A. $0 \le x \le 300$

B. $0 \le x \le 20$

C. $0 \le y \le 300$